- The gibbs free energy which is the combination of the above two ($\Delta H - T\Delta S$) should be negative for a reaction to occur spontaneously, i.e. the necessary condition for a reaction to be spontaneous is $\Delta H - T\Delta S < 0$

Table 7.5 Effect of Temperature on Spontaneity of Reactions:

ΔH_r	ΔS_r	$\Delta G_r = \Delta H_r - T\Delta S_r$	Description	Example
–	+	– (at all T)	Spontaneous at all temperature	$2O_3(g) \rightarrow 3O_2(g)$
–	–	– (at low T)	spontaneous at low temperature	Adsorption of gases
		+ (at high T)	non-spontaneous at high temperature	
+	+	+ (at low T)	non-spontaneous at low temperature	Melting of a solid
		– (at high T)	spontaneous at high temperature	
+	–	+ (at all T)	non spontaneous at all temperatures	$2H_2O(g) + O_2(g) \rightarrow 2H_2O_2(l)$

The Table assumes ΔH and ΔS will remain the way indicated for all temperatures. It may not be necessary that way. The spontaneity of a chemical reaction is only the potential for the reaction to proceed as written. The rate of such processes is determined by kinetic factors, outside of thermodynamical prediction.

Problem: 7. 8

Show that the reaction $CO + \left(\frac{1}{2}\right)O_2 \rightarrow CO_2$ at 300K is spontaneous. The standard Gibbs free energies of formation of CO_2 and CO are –394.4 and –137.2 kJ mole^{-1} respectively.

$$CO + \left(\frac{1}{2}\right)O_2 \rightarrow CO_2$$

$$\Delta G^0_{(reaction)} = \sum G^0_{f\,(products)} - \sum G^0_{f\,(reactants)}$$

$$\Delta G^0_{(reaction)} = \left[G^0_{CO_2}\right] - \left[G^0_{CO} + \frac{1}{2}G^0_{O_2}\right]$$

$$\Delta G^0_{(reaction)} = -394.4 + [137.2 + 0]$$

$$\Delta G^0_{(reaction)} = -257.2 \text{ kJ mol}^{-1}$$

$\Delta G_{(reaction)}$ of a reaction at a given temperature is negative hence the reaction is spontaneous.

Evaluate Yourself - 8

For a chemical reaction the values of ΔH and ΔS at 300K are – 10 kJ mole^{-1} and –20 J deg^{-1} mole^{-1} respectively. What is the value of ΔG of the reaction? Calculate the ΔG of a reaction at 600 K assuming ΔH and ΔS values are constant. Predict the nature of the reaction.

7.11.2 Relationship between standard free energy change (ΔG^0) and equilibrium constant (K_{eq}):

In a reversible process, the system is in perfect equilibrium with its surroundings at all times. A reversible chemical reaction can proceed in either direction simultaneously, so that a dynamic equilibrium is set up. This means that the reactions in both the directions should proceed with decrease in free energy, which is impossible. It is possible only if at equilibrium, the free energy of a system is minimum. Lets consider a general equilibrium reaction

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